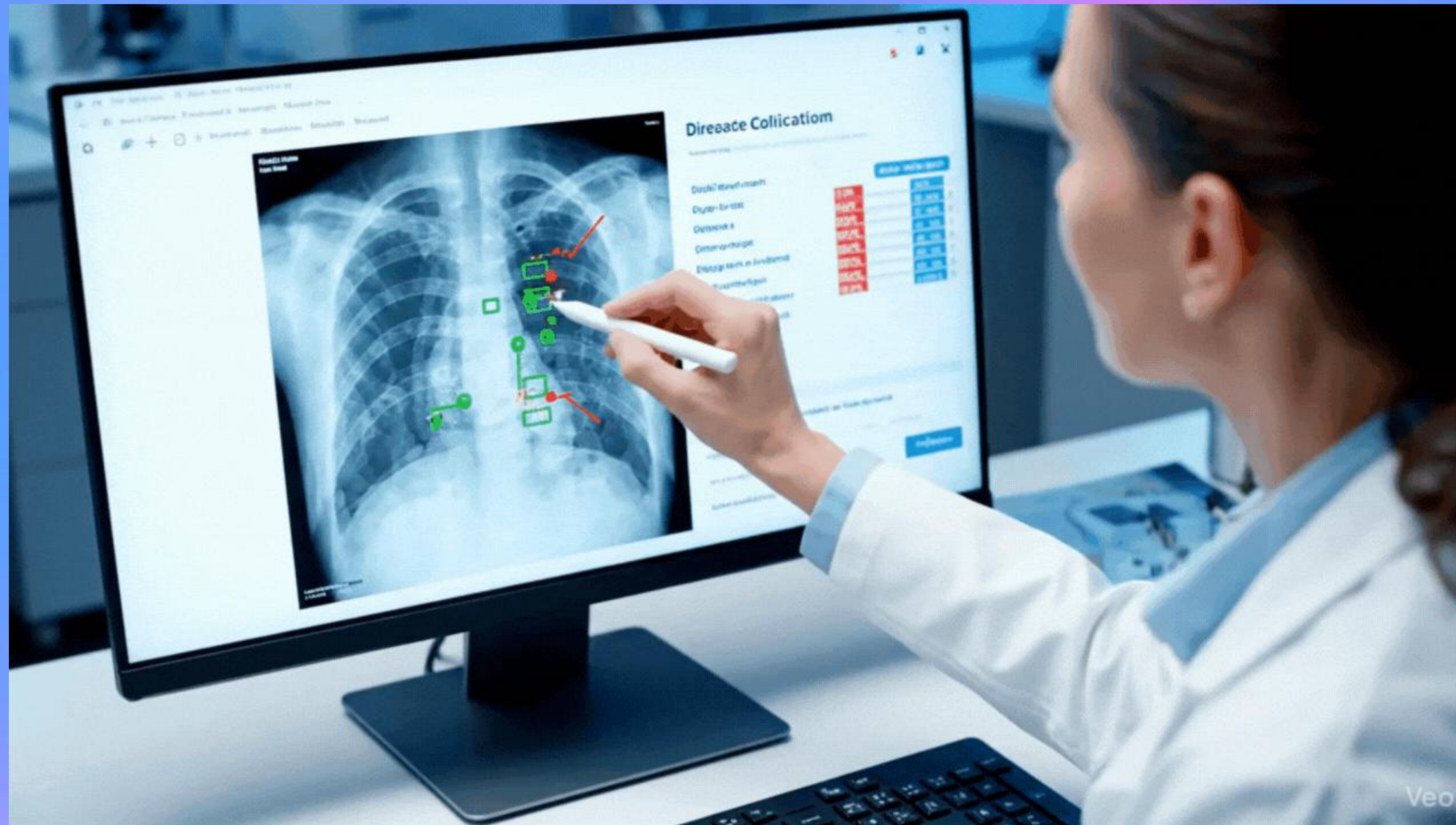


CLINISCAN : LUNG-ABNORMALITY DETECTION ON CHEST X-RAYS USING AI 🦴



- An AI-driven approach for accurate and automated lung abnormality diagnosis

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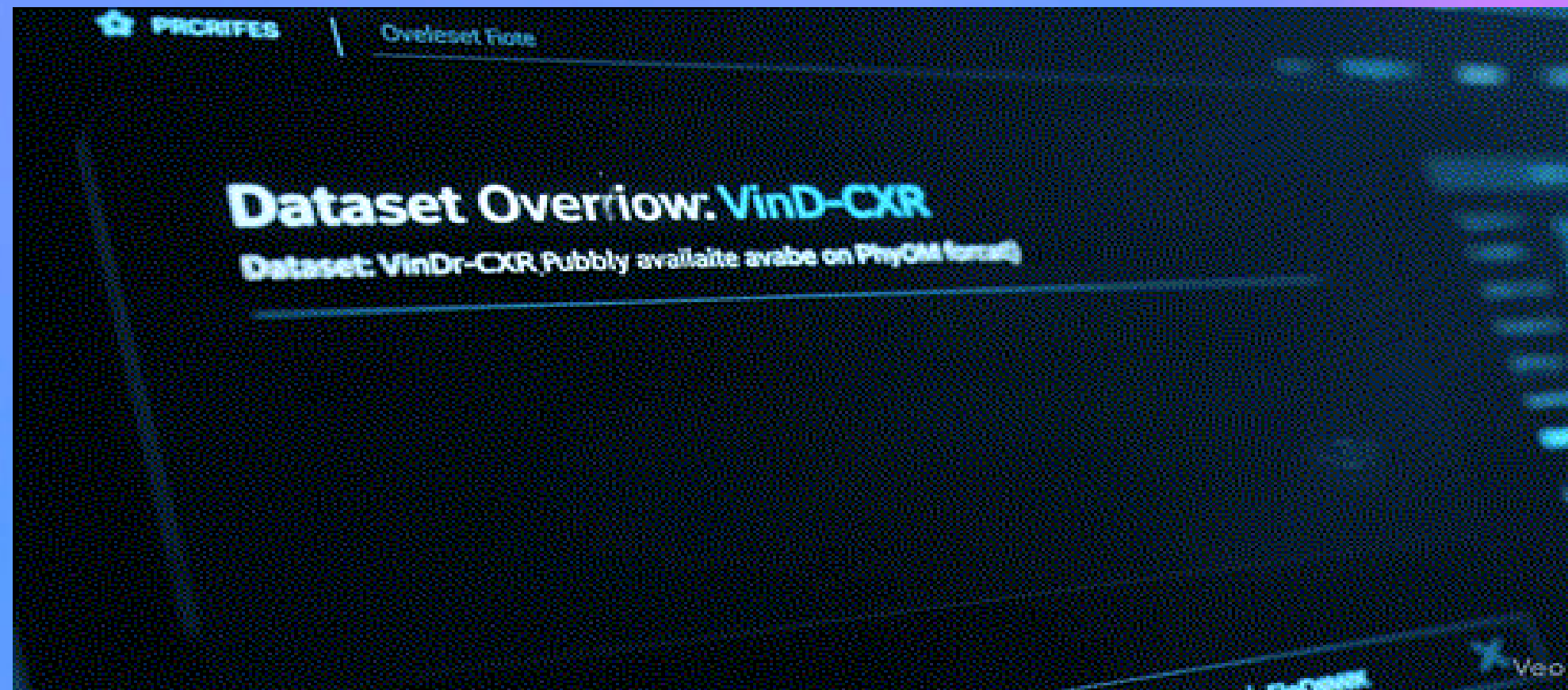
Project Overview

- CLINISCAN is an AI-based project that detects lung problems using chest X-ray images.
- It uses deep learning to find signs of diseases like pneumonia or tuberculosis.
- The system helps doctors check X-rays quickly and make accurate decisions.
- It aims to improve early detection and support better healthcare.

Dataset Overview

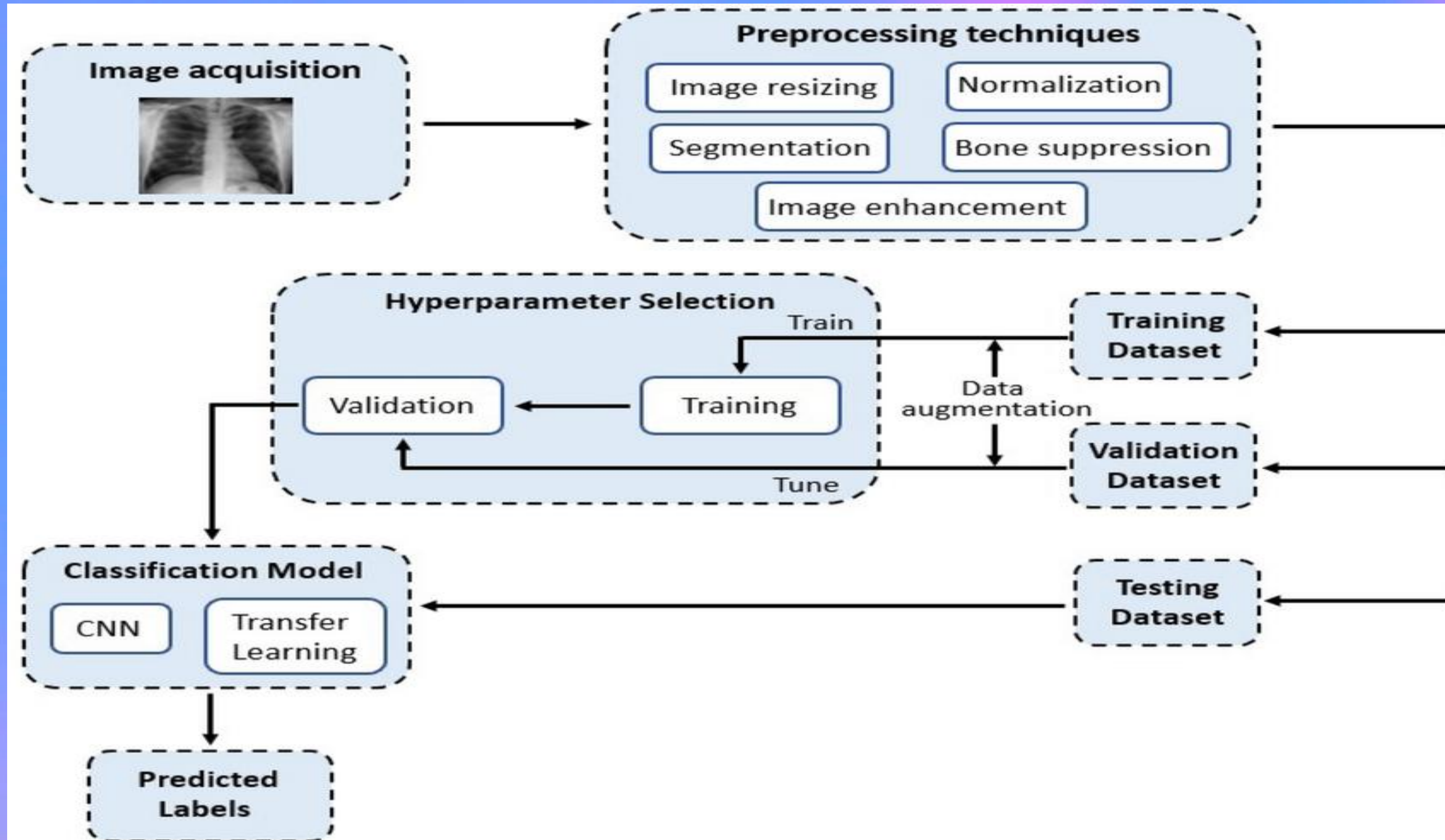
- The project uses chest X-ray images from publicly available datasets like **Kaggle**.
- Contains **~100,000** X-ray images labeled for various lung conditions.
- Labels include **Normal, Pneumonia, Tuberculosis, Lung Opacity**, etc.
- Images are **preprocessed**: resized, normalized, and augmented for training.
- The dataset is split into **training, validation, and test sets** for model evaluation.

Reason for Choosing Dataset



- Real-world dataset from **trusted medical sources**.
- Covers **multiple lung conditions**, allowing accurate classification.
- Well-annotated and suitable for **AI research**.
- Provides **enough images** for deep learning model training.
- Helps build a system that can be **useful in real clinical scenarios**

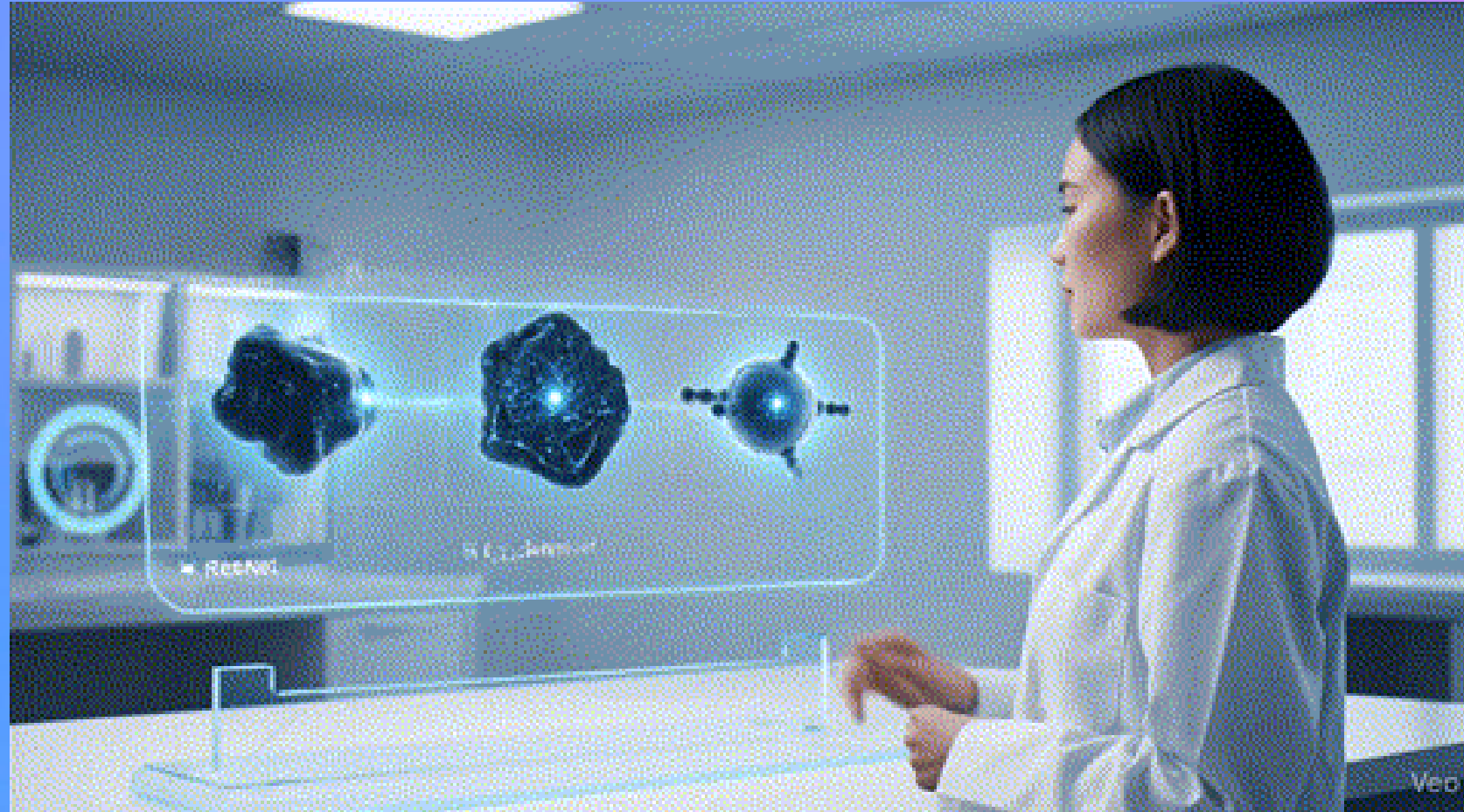
Methodology



Methodology

- **Image Acquisition:** Chest X-ray images are collected.
- **Data Preprocessing:** Convert DICOM → PNG/JPEG, Images are resized and normalized
- **Annotation Conversion:** CSV → YOLO/COCO format.
- **Model Training:** A deep learning model is trained on the preprocessed images.
- **Evaluation:** The model's performance is assessed using metrics like accuracy, precision, and recall.
- **visualization:** Generate Grad-CAM and bounding box overlays.
- **Deployment:** Build web app using Streamlit.

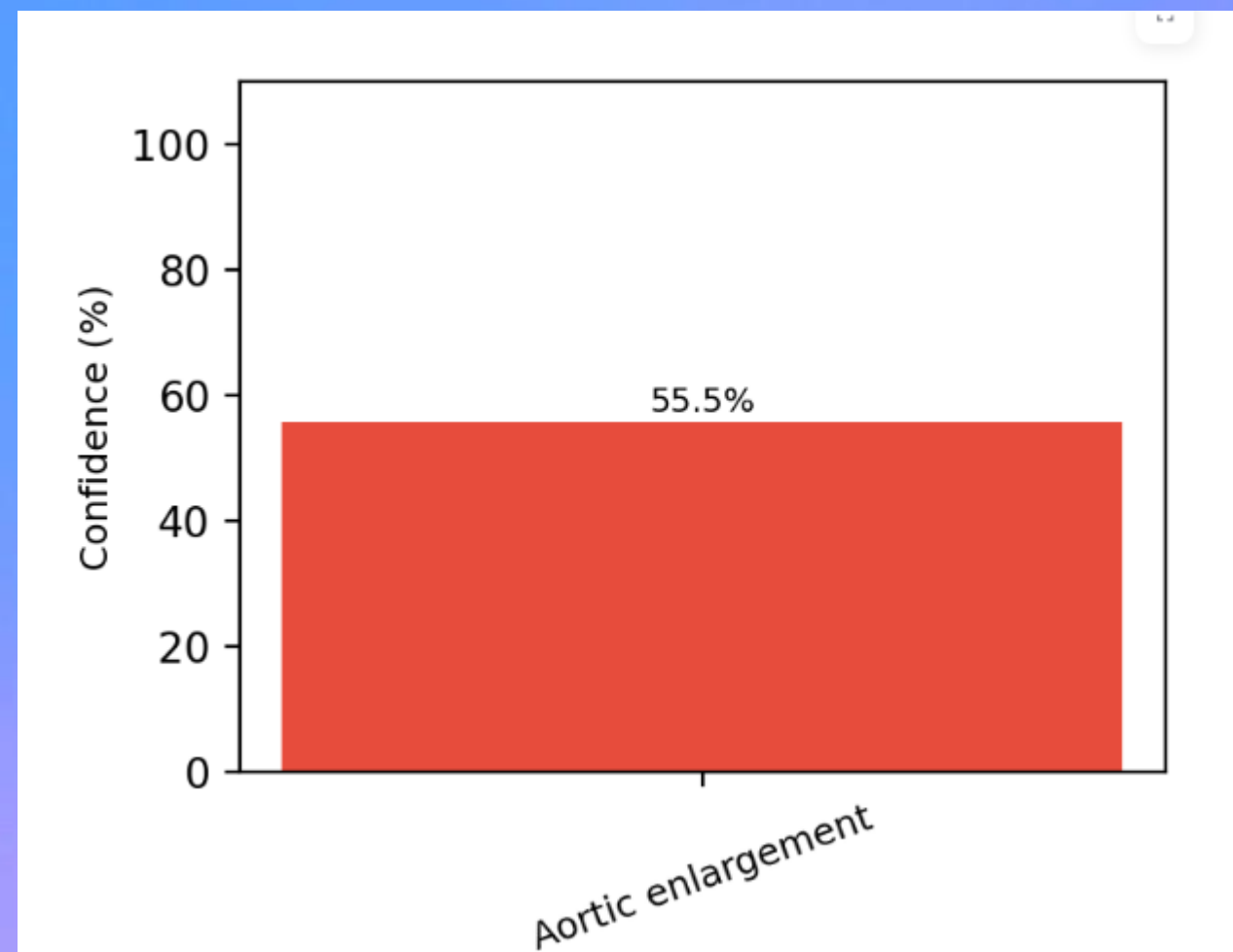
Model Explanation



- **Models Used:** ResNet, EfficientNet, DenseNet, and YOLOv.
- **Purpose:** To detect and classify lung diseases from chest X-ray images.
- **ResNet, EfficientNet, and DenseNet:** Used for image classification and feature extraction.
- **YOLOv Model:** Used for object detection and highlighting the affected lung regions.
- **Output:** Produces probability scores or binary labels (Normal / Abnormal).
- **Advantages:** High accuracy, faster detection, and better visual interpretability.

Training & Evaluation

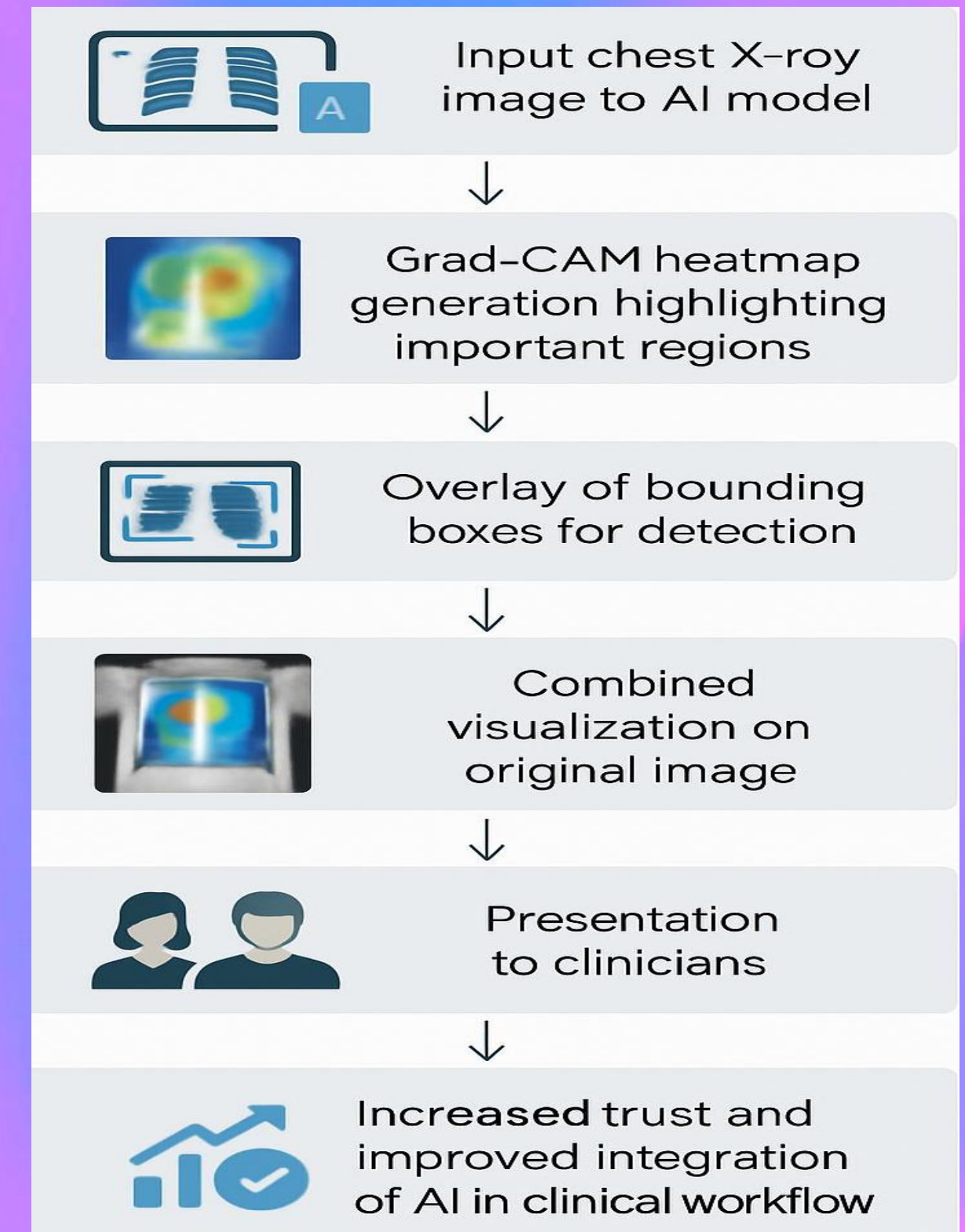
- Training performed using **GPU** (and sometimes **CPU**) for faster computation.
- **Epochs:** 30–50 **Batch Size:** 32 **Learning Rate:** 0.001
- Model performance evaluated using **Accuracy**, **Precision**, **Recall**, and **F1-Score**.
- A **confidence score** is generated to quantify the model's certainty in its predictions.
- The trained model achieved **~90–95% accuracy** in detecting lung abnormalities.



- Confidence score of an sample image

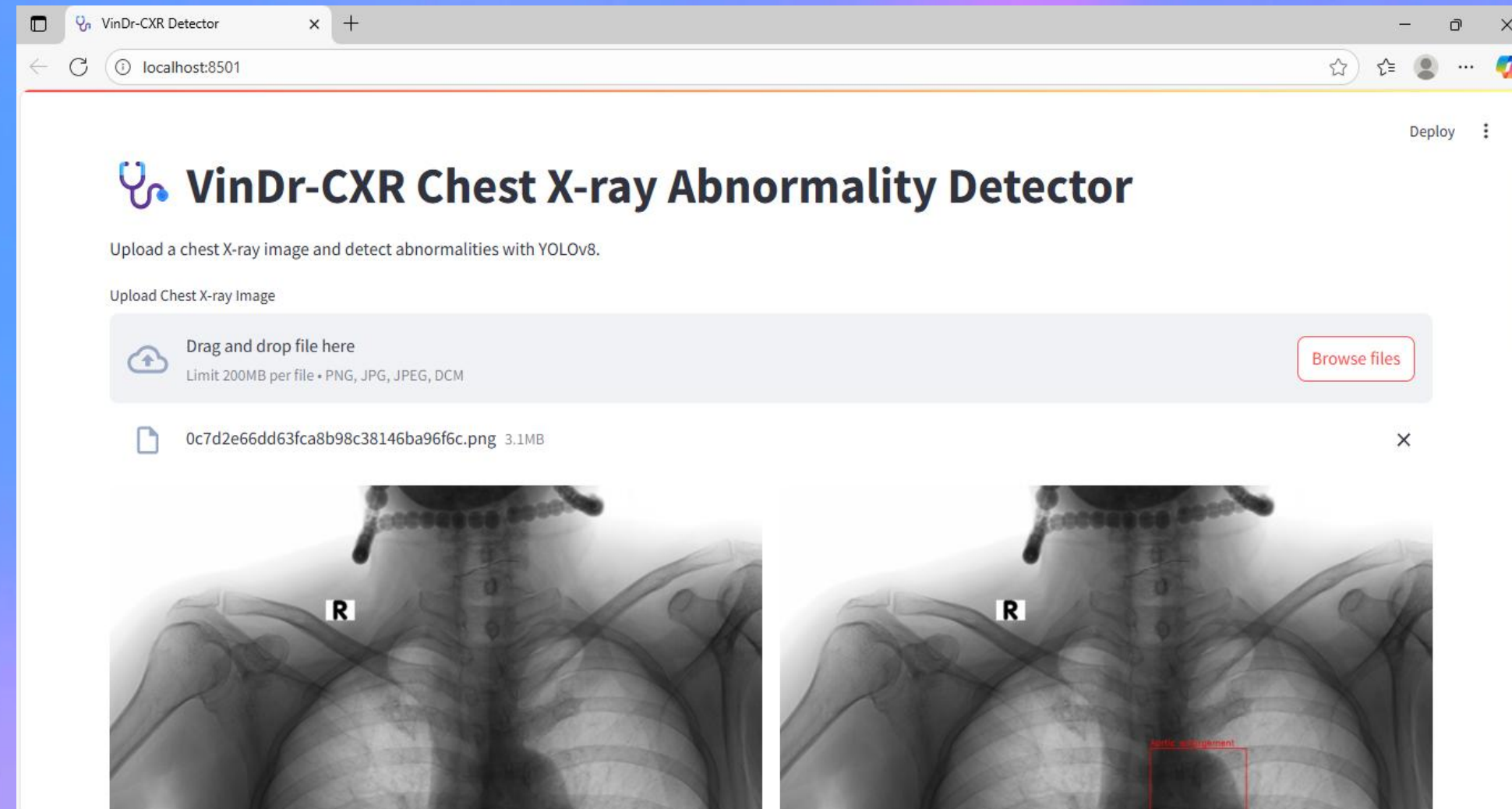
Visualization & Interpretation

- Grad-CAM: Highlights image regions influencing AI predictions.
- Bounding Boxes: Show detected abnormal regions in lungs.
- Combined view increases transparency and clinical trust.
- Helps radiologists understand AI's decision-making process.



User Interface

- Built using **Streamlit**
- Allows users to **upload chest X-ray images** for analysis.
- Provides **real-time prediction**: Normal or Abnormal.
- Displays **heatmaps** highlighting affected areas.
- Simple, interactive design for **doctors and healthcare staff**



Output Demo



Select Mode

Choose analysis type

☒ Classification

☐ Detection

Confidence Threshold

Select minimum confidence

0.50

About the Model

This application uses a ResNet50-based multi-label classifier trained to identify multiple lung

CliniScan: Lung-Abnormality Analysis

Upload a chest X-ray image

 Drag and drop file here

Limit 200MB per file • JPG, JPEG, PNG

Browse files

Share









< Manage app

Future Scope

- Integrate with **hospital management systems (HMS)**.
- Extend to **detect multiple lung diseases** simultaneously.
- Use **3D imaging like CT scans** for higher accuracy.
- Deploy on **mobile and edge devices** for quick diagnosis.
- Explore **privacy-preserving techniques** like federated learning.

Conclusion

- CLINISCAN demonstrates **AI-assisted lung disease detection** using chest X-rays.
- The model achieves **high accuracy** and highlights affected areas for interpretation.
- Supports doctors in **faster, more reliable diagnosis**.
- Improves **early detection and healthcare efficiency**.
- Bridges the gap between **technology and medical care** for better patient outcomes.

Thank you

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